ALGEBRA 1 · UNIT 1

Section 1.1
Multiple Representations of Relationships

A table, a graph, an equation, and a story can describe the same thing. The skill is knowing how to read and connect them.

> **Mathematical idea:** A representation is not just a picture or a rule. It is evidence about how two quantities are connected — and evidence can be checked, challenged, and compared.

Today we will practice identifying representations, deciding what evidence they provide, and judging whether multiple representations describe the same relationship.

Learning Target

Students will connect tables, graphs, equations, and contexts that represent the same relationship.

I can statements

- I can identify when different representations describe the same relationship.
- I can match tables, graphs, equations, and situations.
- I can describe how two quantities are related.
- I can determine what quantities are changing.
- I can justify how representations are connected.

What's To Come

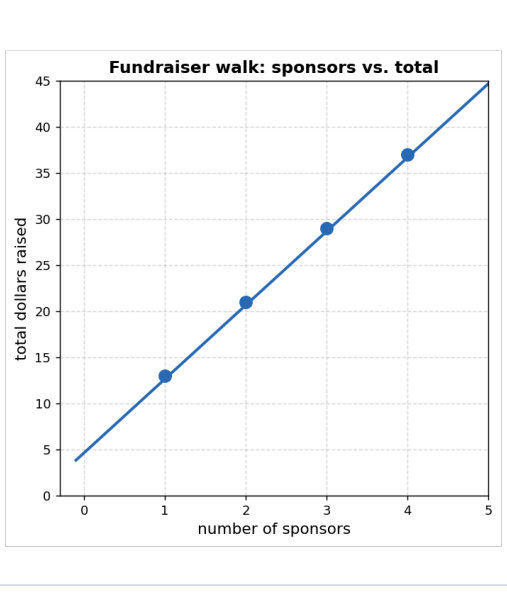
This is a long-range preview for future lessons. Do not solve it today. Instead, annotate it individually. Circle anything familiar, underline symbols or directions that seem important, and write notes about what information you think will be needed later.

As you annotate, ask yourself: Do all four parts seem to tell the same story? Where do you see a possible conflict? What would you need to check to be sure?

Future Task — Representations in Conflict

A school is organizing a charity walkathon. A student creates four representations to track the total amount raised. Another student says, "Something doesn't match. I can't tell which one is wrong."

Graph



Table

Sponsors (s)	1	2	3	4
Total raised (T)	13	21	29	37

Equation

$$T = 8s + 4$$

Context

Each sponsor pledges $8. There is also a one-time registration fee of $5.

1. Circle any part that looks familiar to you.
2. Underline one piece of evidence that might reveal where the conflict is.
3. Write one question you would need to answer before deciding which representation is wrong.

Intro

Directions

Read the introduction aloud with a partner or in a small group. As you read, annotate the text by highlighting important ideas, circling key vocabulary, and writing short notes about connections, questions, or predictions in the margins.

A relationship describes how two quantities are connected. For example, the total cost of tickets is connected to the number of tickets bought. The distance a person walks is connected to the time spent walking. In each case, one quantity changes, and the other quantity changes along with it.

Mathematicians use different representations because each one makes certain information easier to see. A table shows specific values and ordered pairs. A graph shows the overall shape and pattern. An equation shows a general rule that can generate new values. A context explains what the numbers actually mean in a real situation.

No single representation tells the whole story. A graph shows a pattern but may not show what the variables mean. A context explains the situation but may not show exact values clearly. An equation is efficient but can feel disconnected until we interpret each part.

When we decide whether representations match, we use evidence. We check whether the same ordered pairs appear, whether the starting value is the same, whether the rate of change is consistent, and whether labels, units, and meaning are aligned. A match is not based on how things look at first glance — it is based on what the mathematics actually shows.

Stop and Discuss

1. Why might it be useful to check more than one representation before deciding they match?
2. What could go wrong if someone matches representations by looking at shape or direction only?
3. What kind of error would be hardest to notice just from looking at a graph?

Representations Tell Different Parts of the Story

MATH NOTES **Four Common Representations**

A relationship between two quantities can often be shown in several ways. Each representation answers a different kind of question.

Table

Shows specific input-output pairs. Useful for checking exact values and reading the starting value and rate of change directly.

Graph

Shows the visual pattern across the whole relationship. Useful for seeing trends, comparing quantities, and reading approximate values.

Equation

Shows a rule that works for any input value. Useful when we need to calculate values not already in a table or visible on a graph.

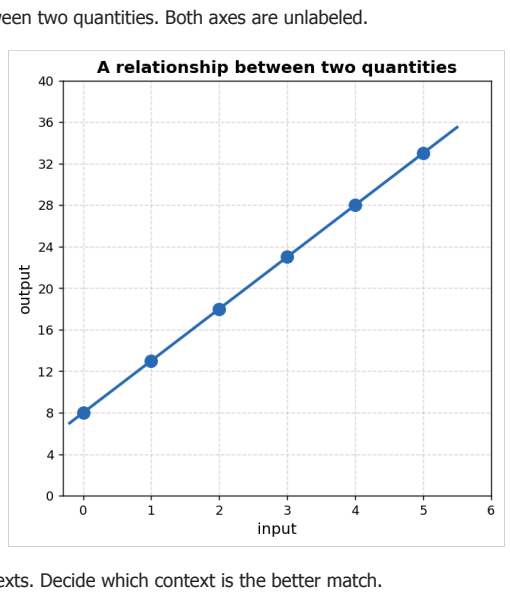
Context

Shows what the numbers actually mean. Useful for interpreting units, understanding variables, and deciding whether an answer makes sense.

YOUR NOTES

Example 1: Match a Graph to a Context

The graph shows a relationship between two quantities. Both axes are unlabeled.



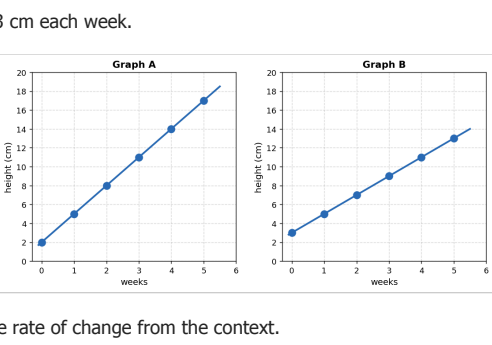
Two students propose different contexts. Decide which context is the better match.

1. A rewards program starts a member with 8 bonus points and awards 5 points per purchase.
2. A rewards program starts a member with 8 bonus points and awards 2 points per purchase.

Both contexts have the same starting value. What evidence must you use to tell them apart? Name the specific part of the graph that provides that evidence.

You Try It 1: Which Graph Matches the Context?

A plant starts at 2 cm tall and grows 3 cm each week.



1. Identify the starting value and the rate of change from the context.
2. For each graph, read the starting value and estimate the rate of change from the graph.
3. Which graph, A or B, matches the context? Use both the starting value and the rate of change as evidence. Explain why the other graph does not match.

MATH NOTES **Evidence That Representations Match**

When comparing representations, do not rely on a quick impression or the overall direction. Look for mathematical evidence.

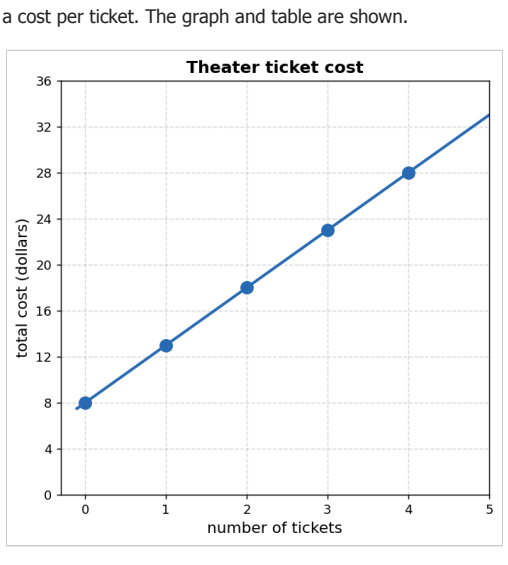
- **Same ordered pairs:** A point from a table should appear on the graph or satisfy the equation.
- **Same starting value:** What happens when the input is 0? Check all representations.
- **Same rate of change:** Does the output change by the same amount each time the input increases by 1?
- **Same meaning:** Do the variables, labels, and units make sense together in context?

Evidence sentence frame: These representations match because __________, and another piece of evidence is __________.

YOUR NOTES

Example 2: Cross-Check a Graph and Table

A theater charges a starting fee plus a cost per ticket. The graph and table are shown.



Tickets (t)	0	1	2	3	4
Total cost (C)	8	13	19	23	28

1. Check at least three ordered pairs from the table against the graph. Do they all appear on the line?
2. If you find a discrepancy, identify which representation you trust more and explain why.
3. Write an equation that best describes the relationship you see in the graph. Explain each part.

You Try It 2: Does the Equation Match the Context?

A delivery service posts this equation: $C = 9m + 3$, where C is total cost in dollars and m is miles driven.

Context: The company charges a flat fee of $3 plus $9 for each mile driven.

1. What does the 9 represent in the equation? What does it represent in the context? Are they playing the same role?
2. Compute the cost for 2 miles using the equation. Then compute it using the context description. Do you get the same value?
3. Do the equation and context match? If not, write the equation that would correctly match the context and explain what you changed.

Stop and Discuss

A student says, "The table and graph both go up, so they must represent the same relationship." What is wrong with that reasoning, and what should the student check instead?

MATH NOTES **Agree, Disagree, or Not Enough Information**

When someone makes a claim about representations, you have three possible responses. Decide which one is supported by the evidence you actually have.

- **Agree** when the mathematical evidence clearly supports the claim across all representations.
- **Disagree** when the evidence contradicts the claim — you can show a specific pair or value that fails.
- **Not enough information** when important labels, values, units, or context are missing and you cannot decide yet.

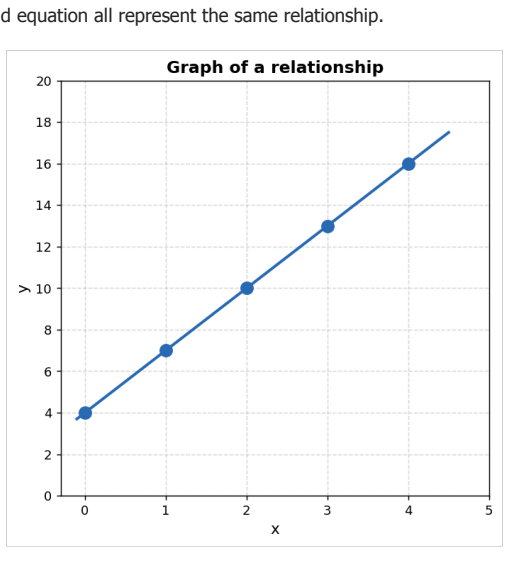
The goal is not to be right quickly. The goal is to name your evidence and say exactly what it shows.

EVIDENCE I SEE

QUESTION I STILL HAVE

Example 3: Find the Error

A student claims the graph, table, and equation all represent the same relationship.



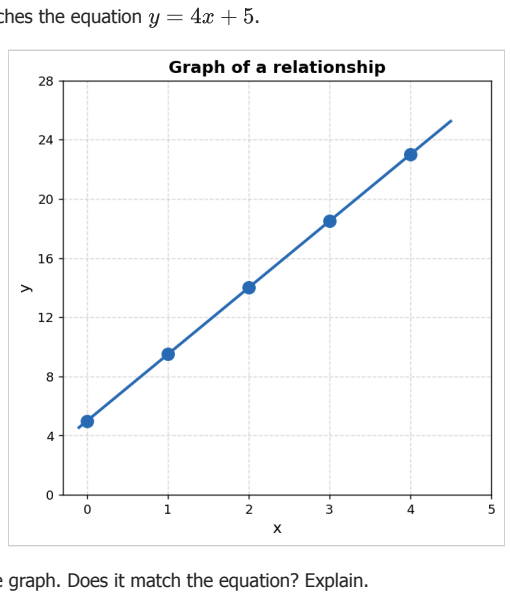
x	0	1	2	3
y	4	7	9	13

Equation: $y = 3x + 4$

1. Check each ordered pair in the table against both the graph and the equation. Which pair is inconsistent?
2. Which representation contains the error? How do you know?
3. What should the correct value be, and what evidence supports your correction?

You Try It 3: Does the Graph Match the Equation?

A student says the graph below matches the equation $y = 4x + 5$.



1. Read the starting value from the graph. Does it match the equation? Explain.
2. Choose two points on the graph and calculate the rate of change. Does the rate match the equation?
3. Do you agree, disagree, or need more information? Use your evidence to defend your answer.

MATH NOTES **Creating a Missing Representation**

When you create a representation that is missing, you are not just copying numbers. You must make decisions that keep the same mathematical relationship.

1. Identify the two changing quantities and decide which is the input and which is the output.
2. Use the starting value and rate of change from what you have to guide the new representation.
3. Check: do specific values from one representation also appear in your new one?
4. Ask: if someone only saw your new representation, could they recover the same relationship?

Check question: Does your new representation tell exactly the same story, or did you accidentally change something?

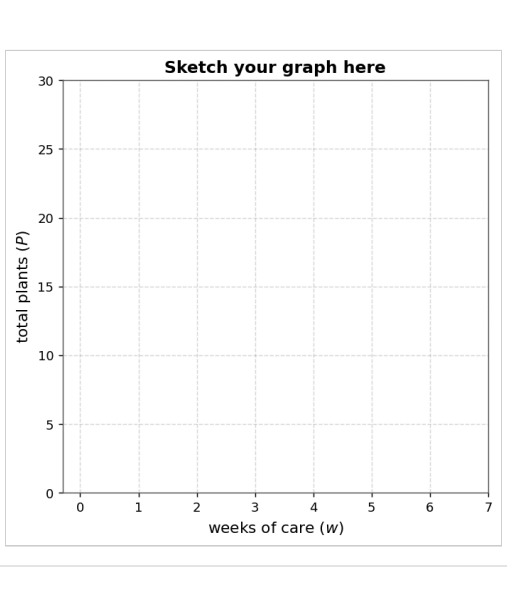
YOUR NOTES

Example 4: Create the Missing Representations

A community garden program starts each plot with 5 plants and adds 4 new plants per week of care.

Weeks of care (w)	0	1	2	3	4
Total plants (P)	5	9	13	17	21

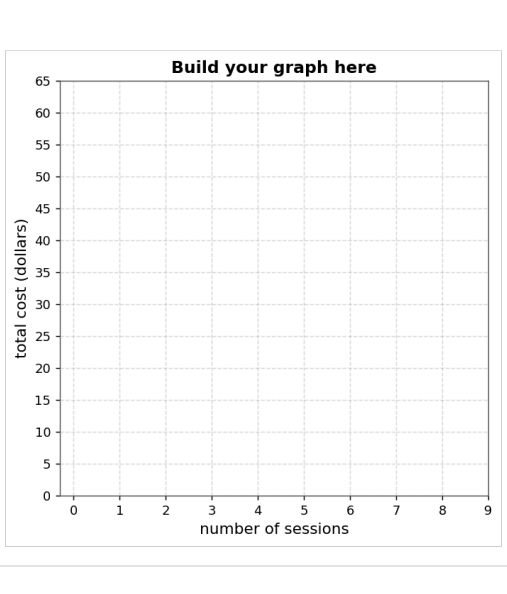
1. Write an equation for total plants P in terms of weeks w. Explain what each part of the equation means in context — do not just name the slope and intercept, explain what they represent about the garden.
2. Plot at least four points from the table on the blank axes below and connect them. Label each point with its coordinates.
3. After you plot, check two things: (a) Does the point where your graph crosses the vertical axis match the starting value in the context? (b) For every additional week, does your graph rise by the correct amount? If either check fails, identify which representation to trust and why.



You Try It 4: Build a Full Representation Set

A tutoring service charges a one-time registration fee of $6 plus $5 per session.

1. Create a table with at least five input-output pairs, starting at 0 sessions.
2. Write an equation for the total cost C in terms of the number of sessions s.
3. Plot the relationship on the blank graph below. Label at least three points.
4. Explain how someone could verify that your table, equation, and graph all describe the same relationship. What specific checks would they perform?



Summary

Tables, graphs, equations, contexts, and verbal descriptions can all represent the same relationship. Each representation reveals different information, so strong mathematical reasoning uses more than one representation as evidence.

To decide whether representations match, check ordered pairs, starting values, rates of change, labels, units, and the meaning of the quantities. A match should be supported by evidence across multiple representations, not just appearance or direction.

Creating a new representation is not just copying — it requires identifying the same starting value, rate of change, and meaning in the new form.

Common Mistakes

- **Matching by direction only.** Seeing that two representations both increase (or both decrease) does not mean they match. Check starting values and rates of change. Why it is tempting: the overall pattern looks similar at a quick glance.
- **Ignoring the starting value.** Two relationships can have the same rate of change but different starting values, making them completely different. Check what happens when the input equals 0.
- **Using only one ordered pair as evidence.** One matching pair is not enough — one pair might coincidentally work in two different relationships. Check multiple values.
- **Forgetting labels and units.** The numbers may match mathematically but the labels or units may make the match meaningless in context. Ask what each variable represents.
- **Trusting the graph's starting point without reading carefully.** A graph may appear to start at a certain value, but if the axis does not start at 0 or the scale is uneven, the starting value can be misread. Check the scale before concluding.

Optional Future Task Reflection

Look back at the future task. Do not solve it yet. Highlight one part that makes more sense now than it did at the beginning. Circle one part that still feels unfamiliar. Write one sentence about what representation you would look at first to begin resolving the conflict.